Praktikum Hochfrequenz-Schaltungstechnik WS25/26

Bericht

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1.5.1

Figure 1 shows the measured resistance R_s and impedance $|Z|$ of a carbon resistor and a coaxial termination over a frequency range between 1 MHz and 30 MHz. The coaxial termination exhibits behavior very close to the ideal expectation. Both the R_s and the $|Z|$ remain nearly constant at $\sim 50\Omega$ over the entire frequency range. Thus it it well impedance matched.

In contrast, the carbon resistor exhibits a clear frequency dependency. The impedance magnitude increases with frequency and exceeds 51 Ω at higher frequencies. This behavior is caused by parasitic inductive effects inherent to the resistor's internal structure and material.

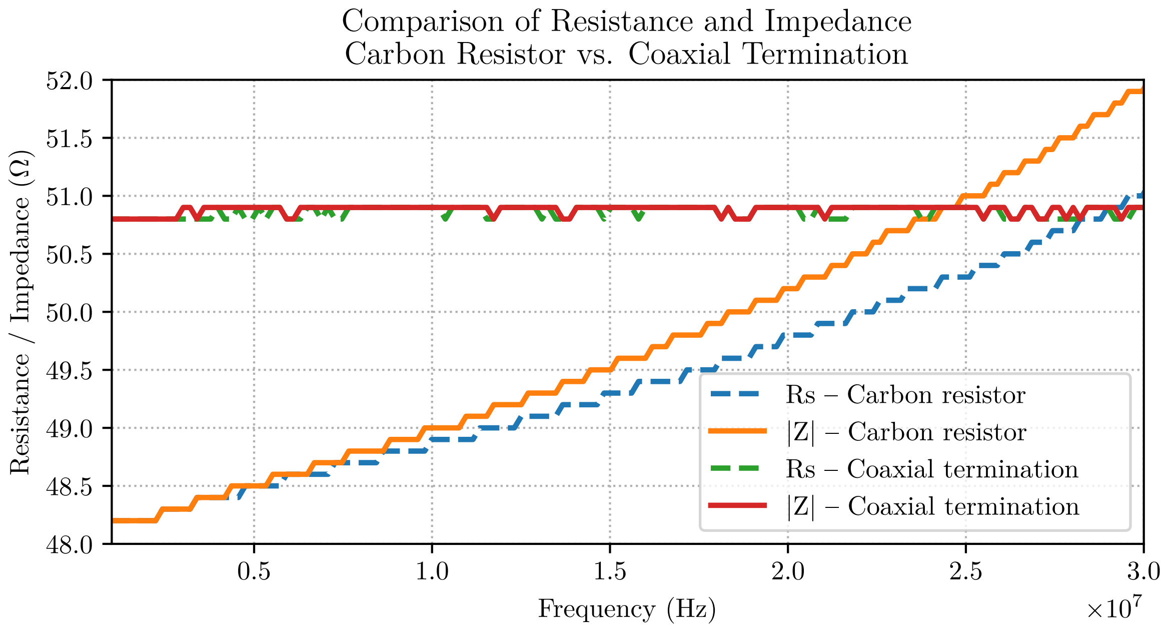

Figure 1: Comparison of resistance and impedance of carbon resistor and coaxial termination

1.5.2

Figure 2 shows the measured impedance magnitude $|Z|$ and phase of the inductor as a function of frequency. At $f = 3.26\,\text{MHz}$, the inductance is measured at $\sim 50\Omega$. The theoretical value is obtained from $|Z| \approx X_L = 2\pi f L \approx 45\Omega$ with L=2.2 μH and is close to the nominal value of the inductor. Small deviations are caused by series resistance and measurement uncertainty. The impedance reaches $|Z| = 50\,\Omega$ at low frequency where the phase is still strongly positive (+88.86°), confirming inductive behavior. A pronounced maximum of $|Z|$ occurs at approximately 30 MHz, indicating the self-resonant frequency of the inductor due to parasitic capacitance. Therefore, the maximum operating frequency of the inductor is below resonance.

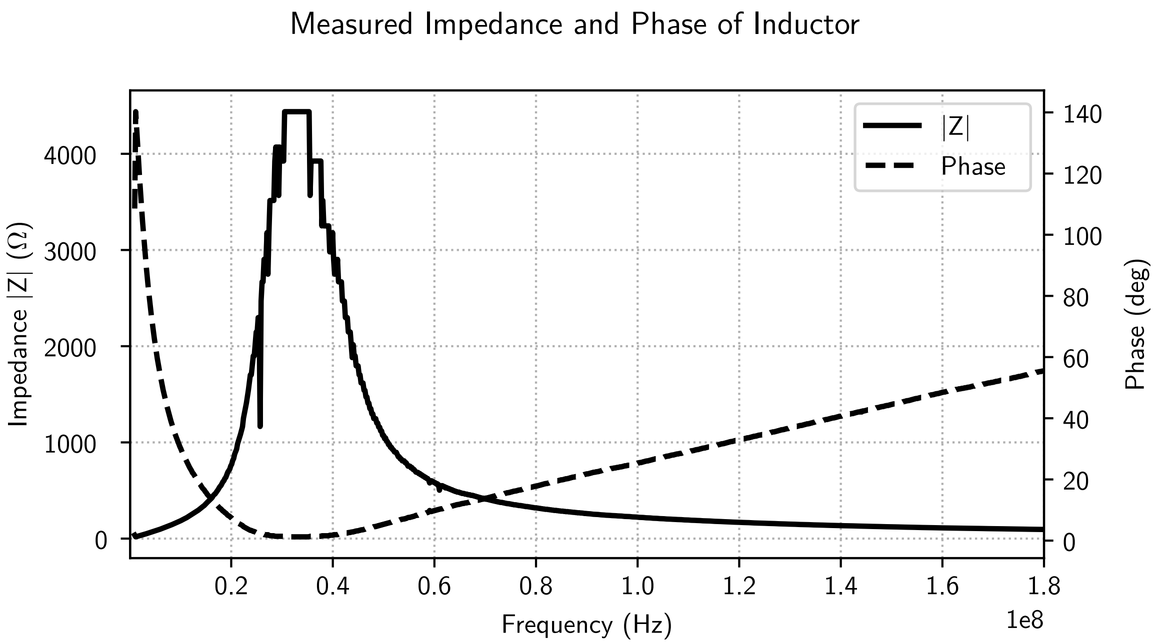

Figure 2: Measured impedance magnitude and phase of an inductor as a function of frequency.

1.5.3

The 3 dB bandwidth is determined from the frequencies where the return loss is 3 dB above the minimum. From the figure, a bandwidth of approximately $\Delta f_{3\text{dB}} \approx 540\,\text{Hz}$ is obtained. The quality factor of the quartz resonator is therefore

$$Q = \frac{f_0}{\Delta f_{3\text{dB}}} \approx \frac{9.8283\,\text{MHz}}{540\,\text{Hz}} \approx 18200.45$$

indicating a very high Q and highly frequency-selective resonator.

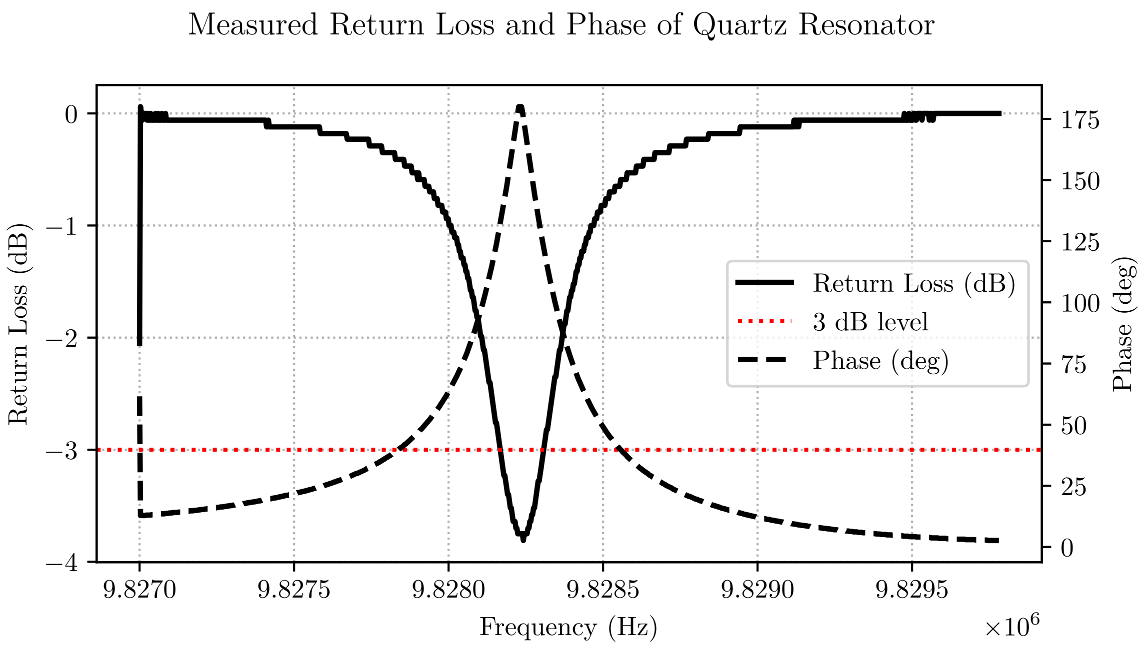

Figure 3: Measured return loss and phase of a quartz resonatoras a function of frequency.

1.5.4

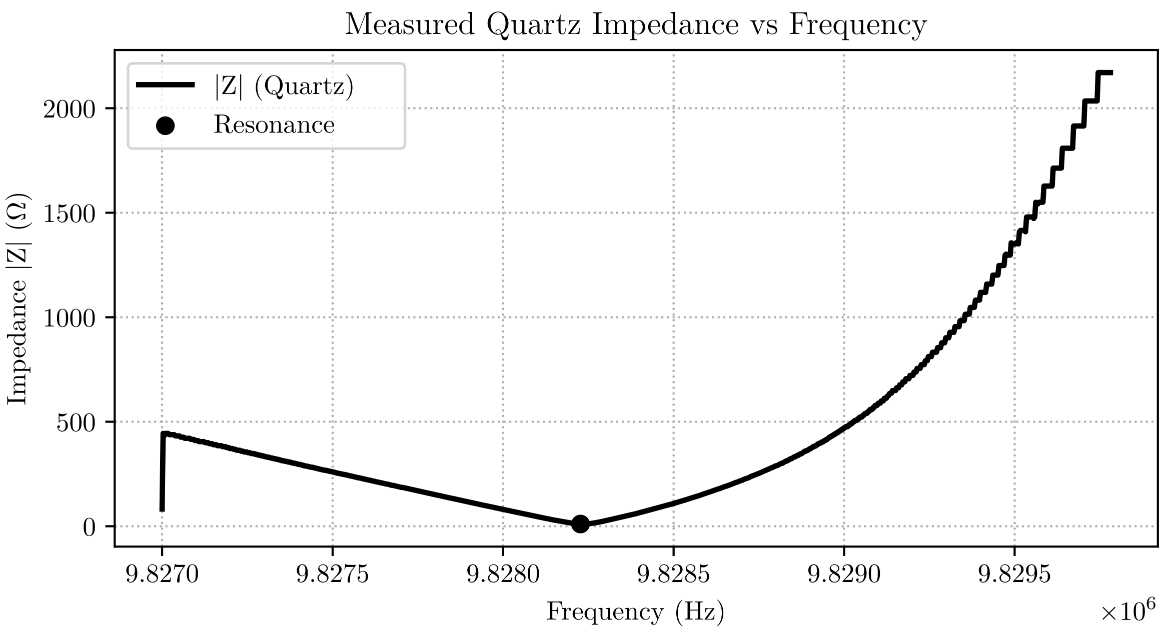

Figure 4: Measured return loss and phase of a quartz resonatoras a function of frequency.

Figure 4 shows the measured impedance magnitude $|Z|$ of the quartz resonator versus frequency. A distinct minimum of the impedance occurs at the resonance frequency $f_0 \approx 9.828227\,\mathrm{MHz}$.

At resonance, the inductive and capacitive reactances of the quartz cancel each other, leaving only the small series resistance. As a result, the impedance reaches its lowest value. Away from resonance, reactive effects dominate, causing the impedance to increase rapidly.

1.5.5

The unknown component is a inductance. This is evident from the impedance behavior as seen in fig. 5. Decreasing at low frequencies and increasing linearly at high frequencies. At the intersection point where $|Z| = Z_0 = 50\,\Omega$, the frequency is $f \approx 1.8 \times 10^7$ Hz.

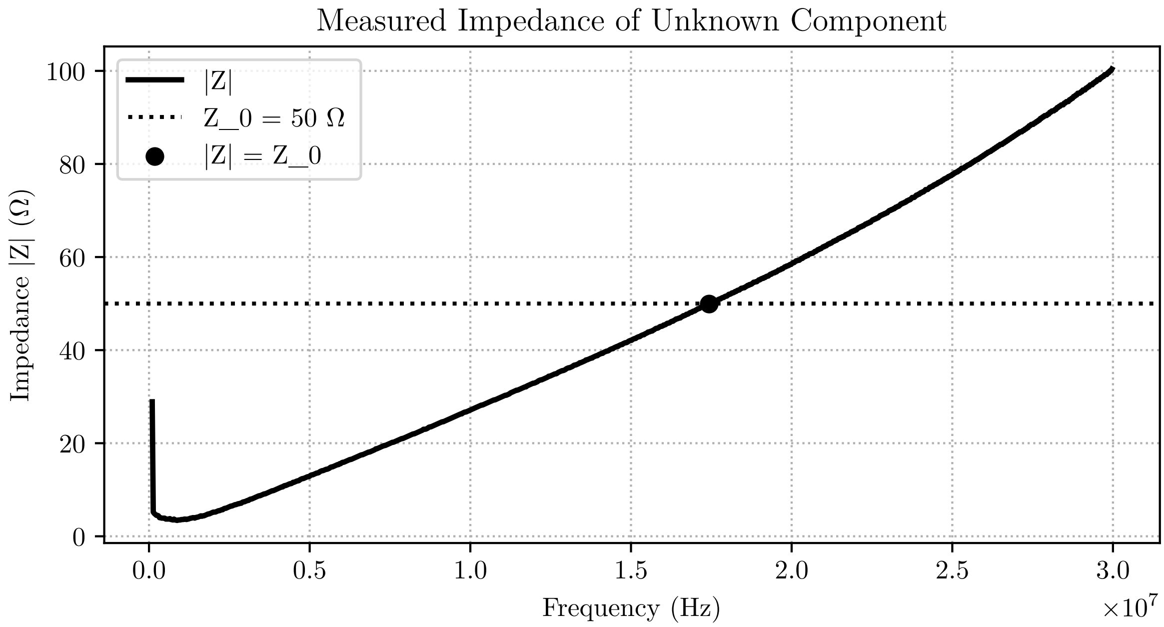

Figure 5: Measured return loss and phase of a quartz resonatoras a function of frequency.

The measured series resistance is 1 Ω. To compare this value to the theoretical value the equation for our reflection at -0.18 dB and 90 degree needs to be evaluated:

$$r = \frac{Z_L - Z_0}{Z_L + Z_0} \Leftrightarrow Z_L = Z_0 \frac{1 + r}{1 - r} \approx 1 + \mathrm{j}\,50\Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{50}{2\pi \cdot 17.434\,\mathrm{MHz}} \approx 456{,}44\,\mathrm{nH}$$

This result matches the measurement very well.

1.5.6

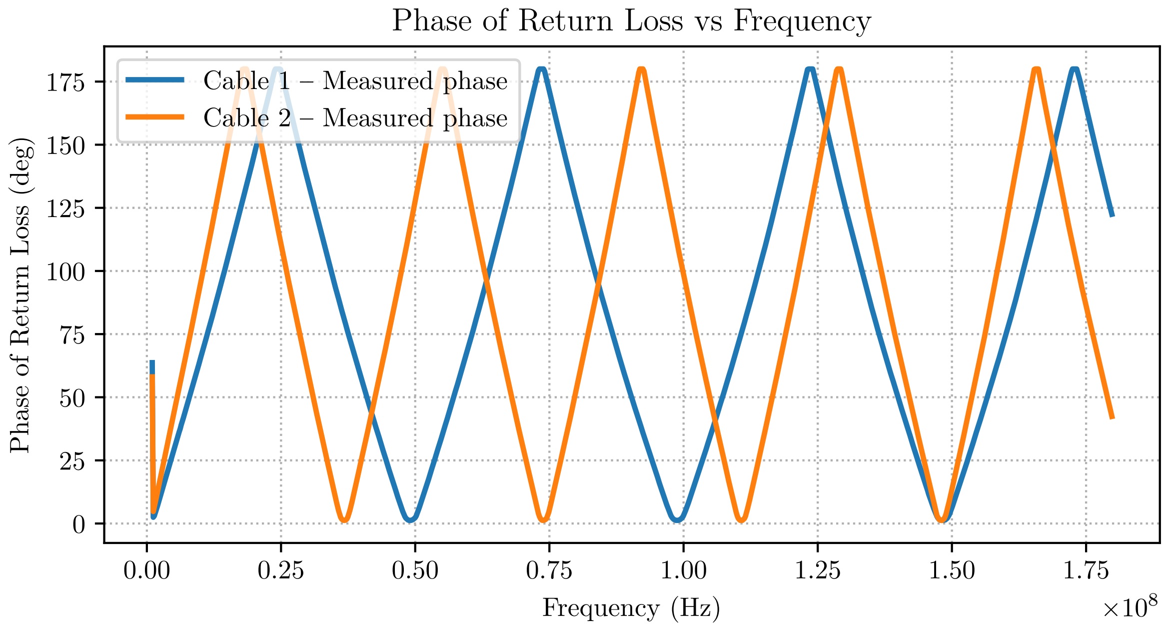

Figure 6: Measured return loss and phase of a quartz resonatoras a function of frequency.

From the formula of the refection coefficient $r = r_L \cdot e^{-j2\beta l}$ and the fact that we can distinctly have maxima at a phase of 180 degree as seen in fig. 6, $-j2\beta l$ is set to π. in addition it is known that $\beta = \frac{2\pi}{\lambda}$ which results in a length of $l = \frac{\lambda}{4}$. After considering a velocity factor of 0.66 and the relation $\lambda_0 = \frac{c_0}{f}$ to connect the frequency to the wave length the resulting equation to calculate the lengths of the cables is $l = \frac{c_0 \cdot VF}{4 \cdot f}$. Cable 1 has the first peak at 24 339 327 Hz leading to a length of 2.7 m. Cable 2 has a peak at 18 359 836 Hz thus a length of 2.03 m.

1.5.7

Figure 7 shows the return loss of a matching network. A resonance can be observed at 14.238 MHz with a loss of -25.51 dB. The SWR is defined as $\frac{1+\rho}{1-\rho}$. With the measured return loss the SWR is 1.11.

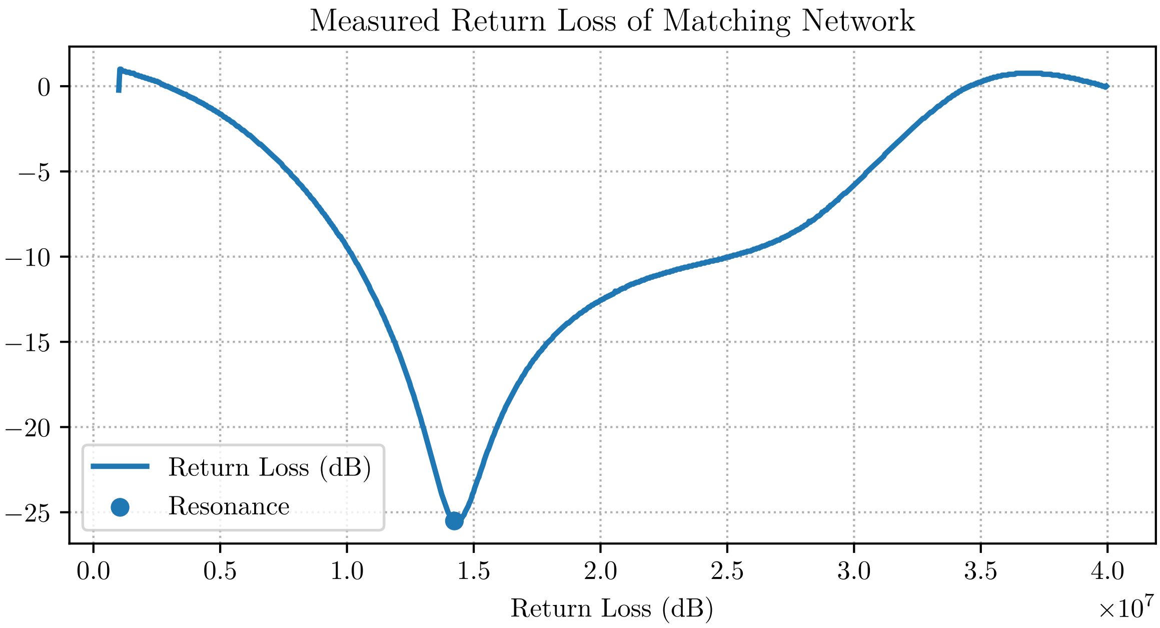

Figure 7: Measured return loss and phase of a quartz resonatoras a function of frequency.

1.5.8

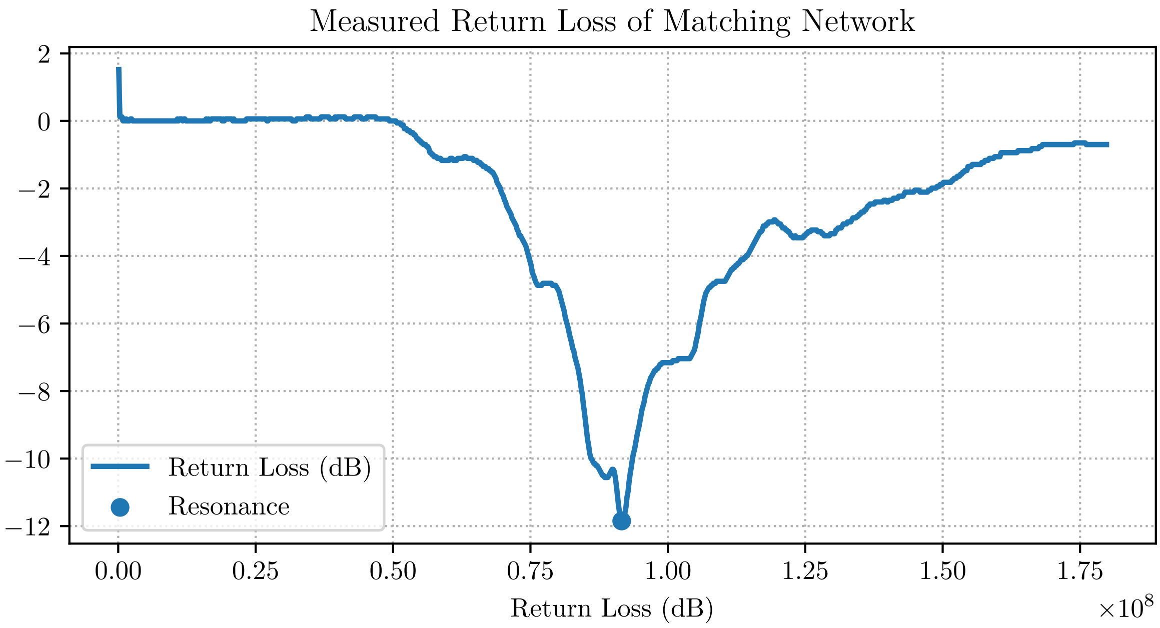

Figure 8: Measured return loss and phase of a quartz resonatoras a function of frequency.

Figure 8 shows a minimum in the reflection coefficient at approximately 91.9 MHz, where the magnitude is measured as -11.85 dB. Substituting this value into the equation below gives the standing-wave ratio (SWR):

$$\mathrm{SWR} = \frac{1 + 10^{\left(\frac{|S_{11}|_{\mathrm{dB}}}{20}\right)}}{1 - 10^{\left(\frac{|S_{11}|_{\mathrm{dB}}}{20}\right)}} = 1.69$$

The impedance can be determined just like shown in chapter 1.5.5, the reflection coefficient phase at resonance was measured at 152.5°. Thus the calculated value results in $Z_L = 30.78 + j7.75\Omega$. The resonance frequency lies within the FM broadcast band (87–108 MHz), the antenna is suitable for use in this frequency range.

1.5.9

For a monopole antenna, the approximate length is given by the quarter-wavelength rule $l = \frac{\lambda}{4}$. Substituting the free-space wavelength relation $\lambda = c_0/f$ yields:

$$l = \frac{c_0}{4f} = \frac{c_0}{4 \cdot 91.601\,\mathrm{MHz}} \approx 0.816\,\mathrm{m} = 0.818\,\mathrm{m}$$

1.5.10

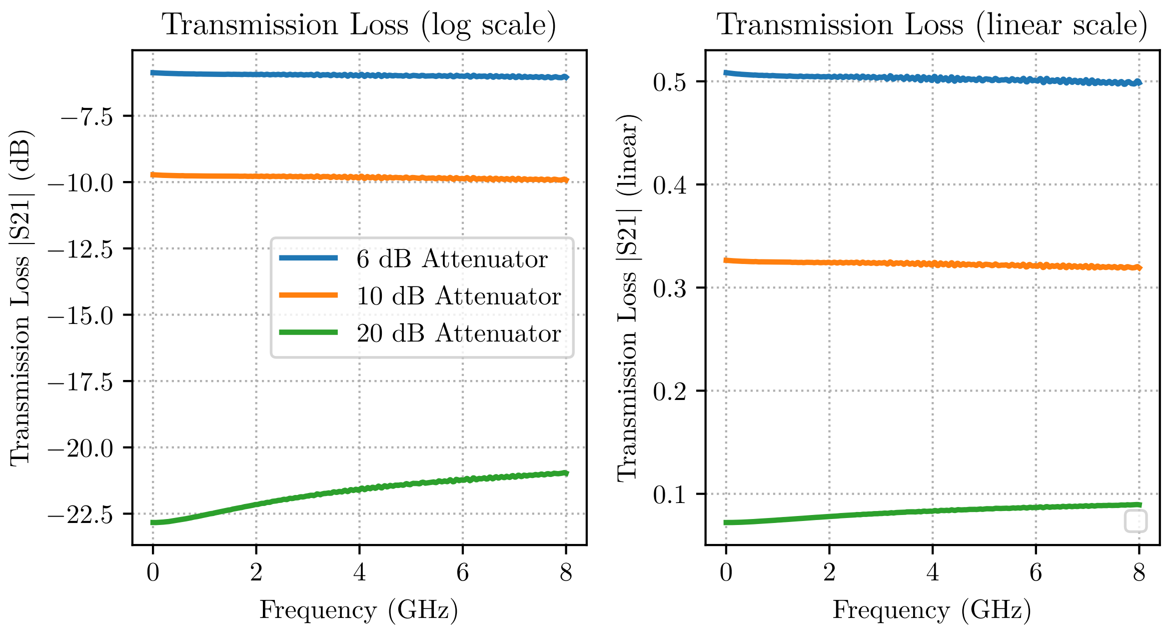

Figure 9: `https://tenor.com/de/view/will-smith-gif-3197096261065537476`

1.5.11

All three attenuators can be used for the entire frequency range of 8 GHz as can be seen in fig. 9. The given cutoff frequency was given on the elements much higher at 18 GHz.

1.5.12

Since the ground creates a nearly perfect reflection due to having a impedance of zero, the signal passes through the attenuator two times resulting in a doubled damping thus a doubled return loss: 12 dB, 20 dB, 40 dB.

1.5.13

In figure 10 the center frequency of the LC band-pass can be determined at 8.502 MHz with a value of -0.18 dB. The return loss is measured at $\sim$ 37 dB. The 3 dB threshold in transmission is measured at 7495021 Hz.

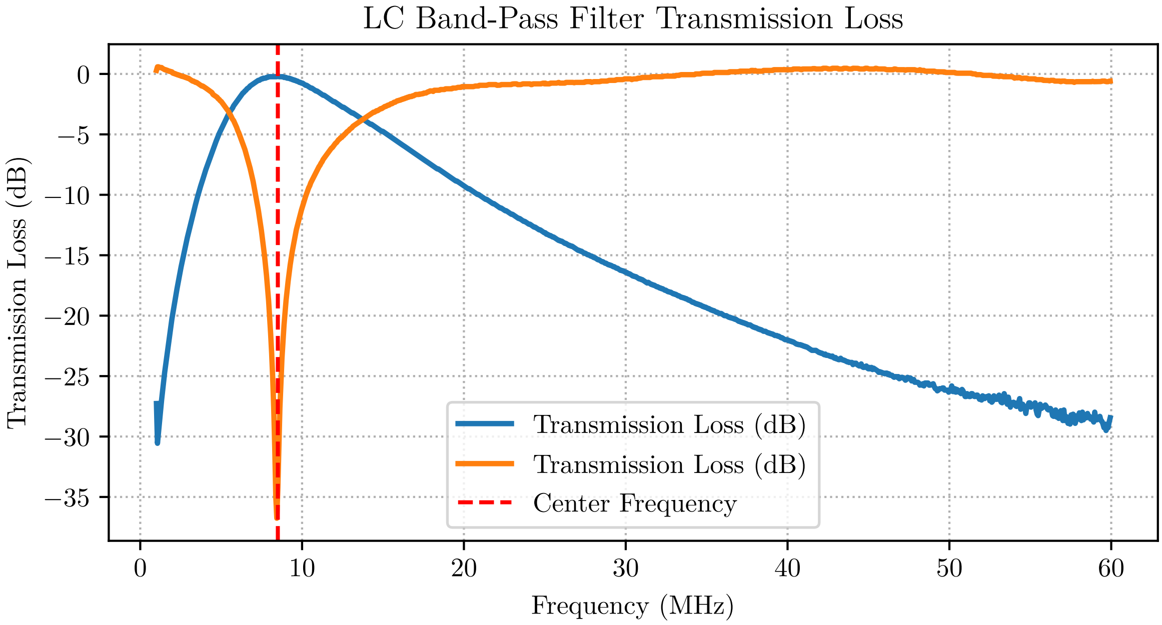

Figure 10: Measured return loss and phase of a quartz resonatoras a function of frequency.

1.5.14

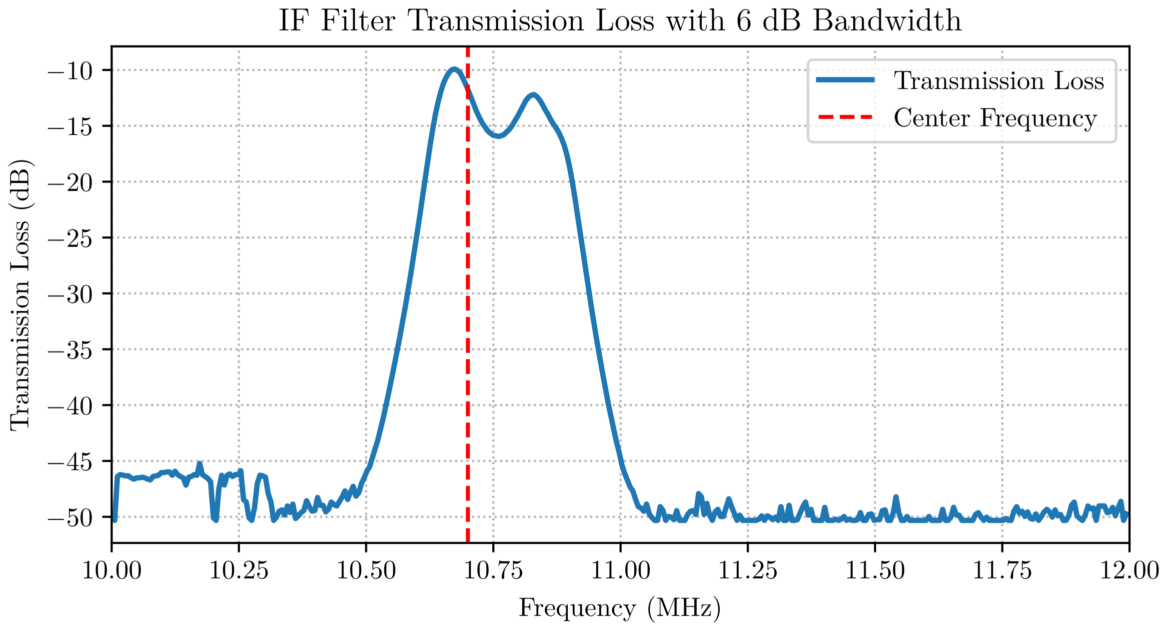

Figure 11: Measured return loss and phase of a quartz resonatoras a function of frequency.

References